DOM Events

Events are signals that something has occurred. (user inputs / actions)

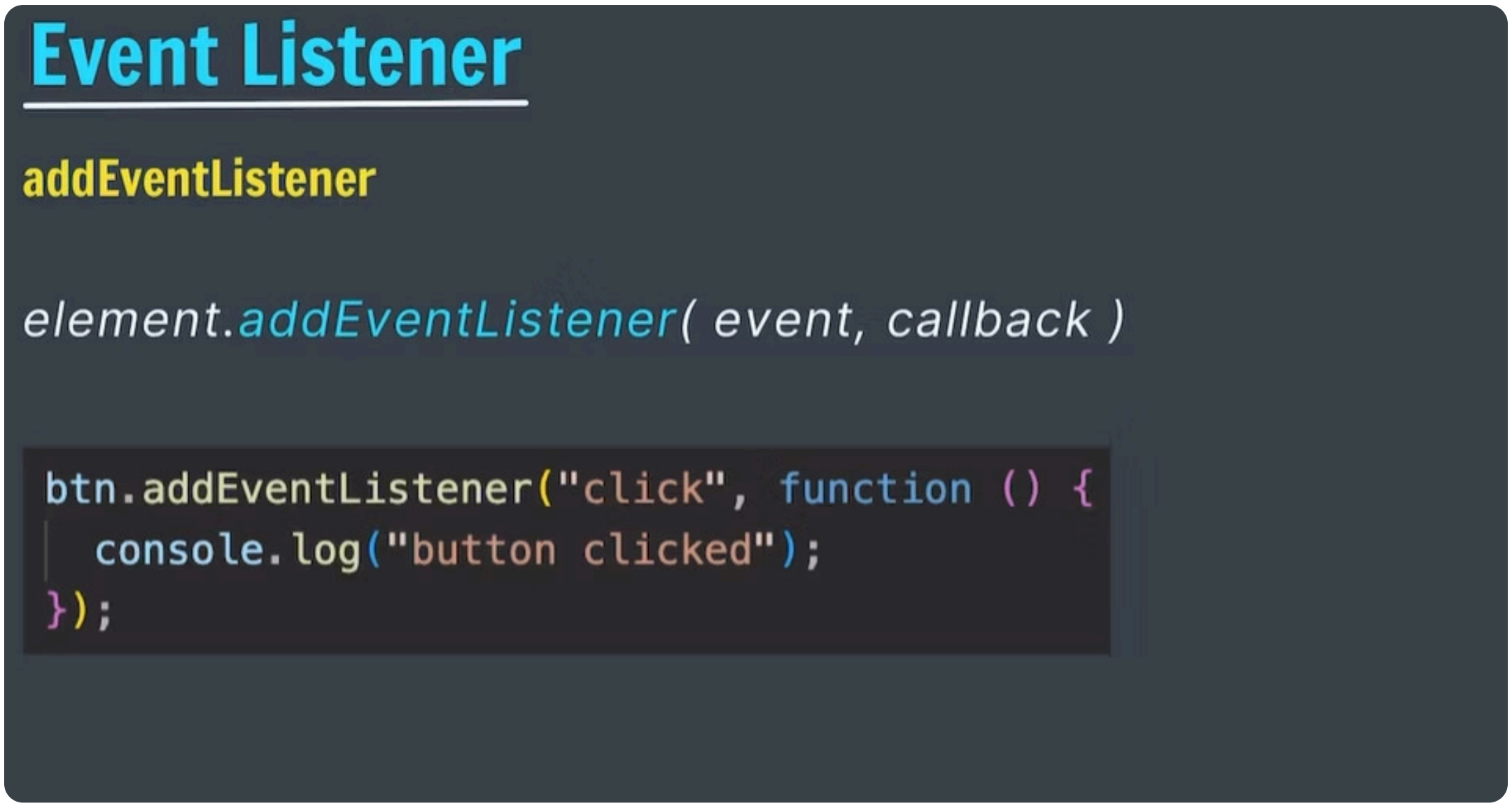
onclick (when an element is clicked)

onmouseenter (when mouse enters an element)

What are DOM Events?

Think of DOM Events as how your web browser and JavaScript talk to each other to make a webpage interactive.

- The **DOM (Document Object Model)** is like a map the browser creates for your webpage, showing all its parts like text, images, and buttons.
- **Events** are actions that happen on this webpage map. These can be things you do, like clicking a button, typing in a search box, or moving your mouse around.
- **JavaScript** is a programming language that can "listen" for these events. When an event happens, JavaScript can jump into action and run specific code.
- This "listening and reacting" is exactly how websites become interactive! It's what makes things happen when you click, type, or interact with a page.



The addEventListener Method

It's the one of the main way we can set"listeners" for events on a webpage element.

Here's how it generally looks:

```
element.addEventListener(event, function)
```

Let's break down each part:

- **element** → This is the specific HTML element you want JavaScript to "watch" for an event. Think of it as the target, like a button, a text input, or even the whole document.
- **event** → This is a string that specifies the type of event you're listening for. Common examples include `"click"` (when a user clicks something), `"keydown"` (when a key is pressed), or `"mouseover"` (when the mouse pointer moves over an element).
- **function** → This is the block of code (also called a "callback function") that will run *automatically* whenever the specified event happens on the chosen element.

Here's a simple example of it in action:

```
const btn = document.querySelector("button");
btn.addEventListener("click", function() {
  alert("Button clicked!");
});
```

Whenever the button is clicked, the function runs automatically, showing an alert!

this in Event Listeners

```
button.addEventListener("click", function () {
  console.log(this); // button element ✅
});

button.addEventListener("click", () => {
  console.log(this); // NOT button ❌ (lexical, often window)
});
```

Keyboard Events

```
let btn = document.querySelector("button");

btn.addEventListener("click", function (event) {
  console.log(event);
  console.log("button clicked");
});

btn.addEventListener("dblclick", function (event) {
  console.log(event);
  console.log("button clicked");
});
```

Form Events

```
let form = document.querySelector("form");
form.addEventListener("submit", function(event){
    alert("form submitted")
})
```

`event.preventDefault()` stops the **default browser action** that normally happens for that event.

Common examples

1) Stop a link from opening

HTML

```
<a href="https://google.com" id="link">Go</a>
<script>
  document.querySelector("#link").addEventListener("click", (e) => {
    e.preventDefault();          // 🚫 won't navigate
    console.log("link clicked");
  });
</script>
```

2) Stop a form from submitting + reloading the page

HTML

```
<form id="f">
  <input name="name" />
  <button>Submit</button>
</form>

<script>
  document.querySelector("#f").addEventListener("submit", (e) => {
    e.preventDefault();          // 🚫 no page refresh
    console.log("form submitted");
  });
</script>
```

3) Stop checkbox default toggle (rare, but possible)

JavaScript

```
checkbox.addEventListener("click", (e) => {
  e.preventDefault(); // checkbox won't change state
});
```

```javascript
let form = document.querySelector("form");

form.addEventListener("submit", function (event) {
  event.preventDefault();

  let user = document.querySelector("#user");
  let pass = document.querySelector("#pass");

  console.log(user.value);
  console.log(pass.value);

  alert(`Hi ${user.value}, your password is set to ${pass.value}`);
});
```

More events

change event

The change event occurs when the value of an element has been changed (only works on <input>, <textarea> and <select> elements).

input event

The input event fires when the value of an <input> , <select> , or <textarea> element has been changed.

```
let form = document.querySelector("form");

form.addEventListener("submit", function (event) {
  event.preventDefault();
});

let user = document.querySelector("#user");

user.addEventListener("change", function () {
  console.log("change event");
  console.log("final value = ", this.value);
});

user.addEventListener("input", function () {
  console.log("input event");
  console.log("final value = ", this.value);
});
```

```
let inp = document.querySelector("#text");
let p = document.querySelector("p");

inp.addEventListener("input", function () {
  console.log(inp.value);
  p.innerText = inp.value;
});
```